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5. Is $S \subseteq T$? Why or why not?

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4. $\forall n \in \mathbb{N}, \exists p$ such that p is prime and $p > n$ *versus* $\exists p, \forall n \in \mathbb{N}$ such that p is prime and $p > n$.

Functions



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4. What changes do we need to make $g(x)$ surjective?

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1. Suppose I have a function $h : S \to T$. Suppose that S has 7 elements and T has 5 elements. Without knowing how h is defined, can we determine if h is injective? Surjective?
2. Using the same sets, what can we determine about $b : T \to S$?

Modular Arithmetic



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2. $32 - 45 \mod 7$
3. $13 \cdot 17 \mod 8$
4. $2 \cdot 3 \cdot 4 \cdot 5 \mod 6$

Putting Two Together



Using modular arithmetic, we can now construct completely new functions using the modular operations! This will be extremely important for nearly every cryptographic scheme we look at.

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$$c(x) = 4x + 1 \mod 5$$

What properties does this function have?

Putting Two Together



Define the function $d : \{0, 1, ..., 10\} \to \{0, 1, ..., 10\}$ by:

$$d(x) = 6^x$$

What properties does this function have?